
Dynamic Modeling and Interaction Analysis of a 2-DOF Aerial Manipulator

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1 Introduction

The demand for mobile robotic platforms capable of dexterous interaction with their environment has given rise to the field of aerial manipulation. Integrating a robotic arm onto an Unmanned Aerial Vehicle (UAV) creates a hybrid system with the reach of a UAV and the manipulation capability of an arm. However, the absence of a fixed base creates a significant control challenge.

This project aims to rigorously model a 2-DOF (degree-of-freedom) robotic manipulator attached to a quadrotor base using MATLAB/Simscape Multibody. Our focus is to quantify the dynamic coupling—the effect where the arm’s motion perturbs the drone’s position, and the drone’s movements affect the arm’s trajectory tracking.

2 Computational Framework and Initial Model

2.1 Physical and Geometric parameters

The simulated system, visualized in the Mechanics Explorer, consists of a central hub (the drone frame) connected to a two-link manipulator arm. The drone is represented by a rigid cross chassis.

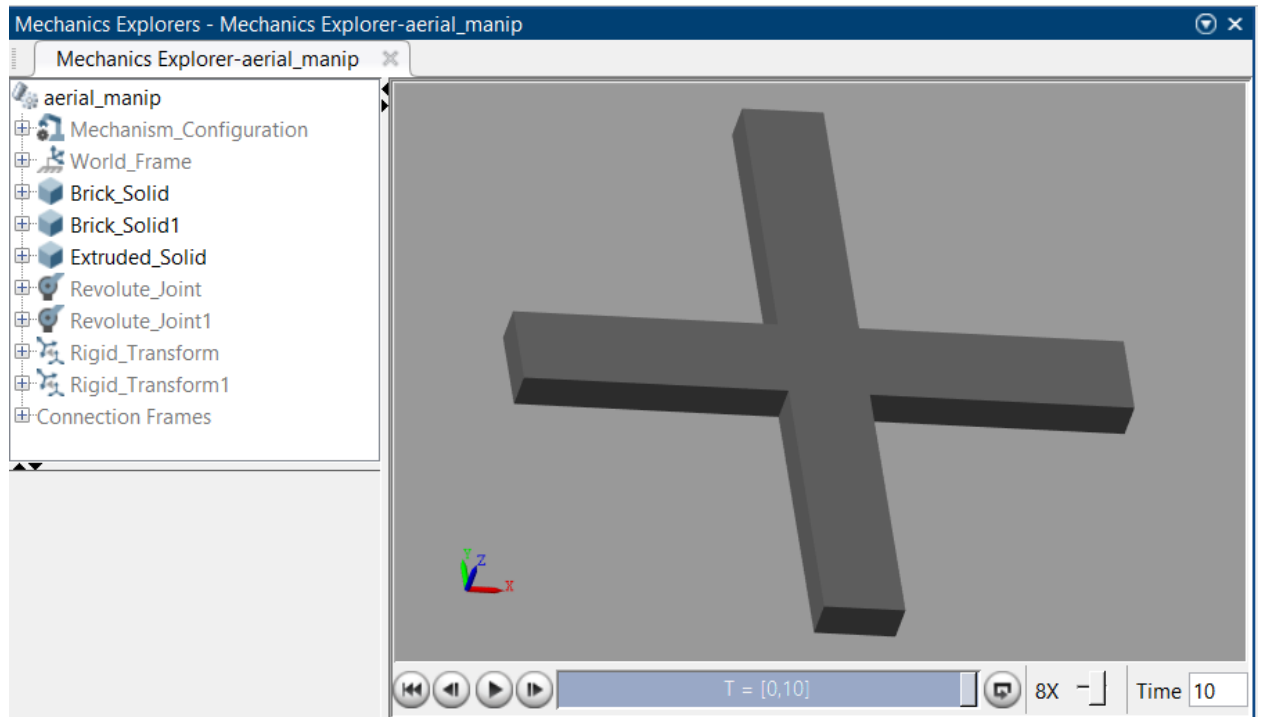


Figure 1: Physical representation of the aerial manipulator frame and links in Simscape Mechanics Explorer.

The physical parameters assumed for numerical computation are crucial for understanding inertial coupling. A primary perspective for engineering robustness is ensuring the drone base is significantly more massive than the attached links to avoid catastrophic instability at the base level.

2.2 Numerical Solvability and Degrees of Freedom

Initially, the simulation encountered specific numerical obstacles during model initialization, as documented by the Diagnostic Viewer in our development stage. Simscape requires a numerically determinate system, meaning every degree of freedom must have defined actuation or constraints.

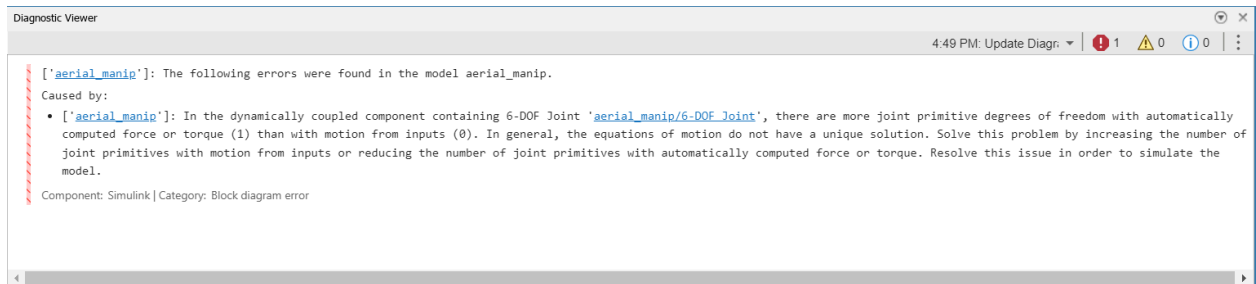


Figure 2: Diagnostic Viewer error indicating a numerically indeterminate state with unconstrained degrees of freedom in the 6-DOF joint.

The error shown in Figure 2 arises because the drone base (the follower of the 6-DOF joint) has 6 unconstrained axes while the arm joints have automatically computed torque.

In general, without numerical damping or joint constraints, the equations of motion are singular. This challenge was resolved by introducing non-zero numerical damping coefficients to all 6-DOF joint primitives to stabilize the solver.

3 System Architecture: Simulink Block Modeling

The modeling methodology utilizes the Simscape Multibody block diagram approach. Figure 3 illustrates the complete system architecture. The 6-DOF Joint provides the essential connection between the static World Frame and the mobile drone body, allowing full translational and rotational freedom.

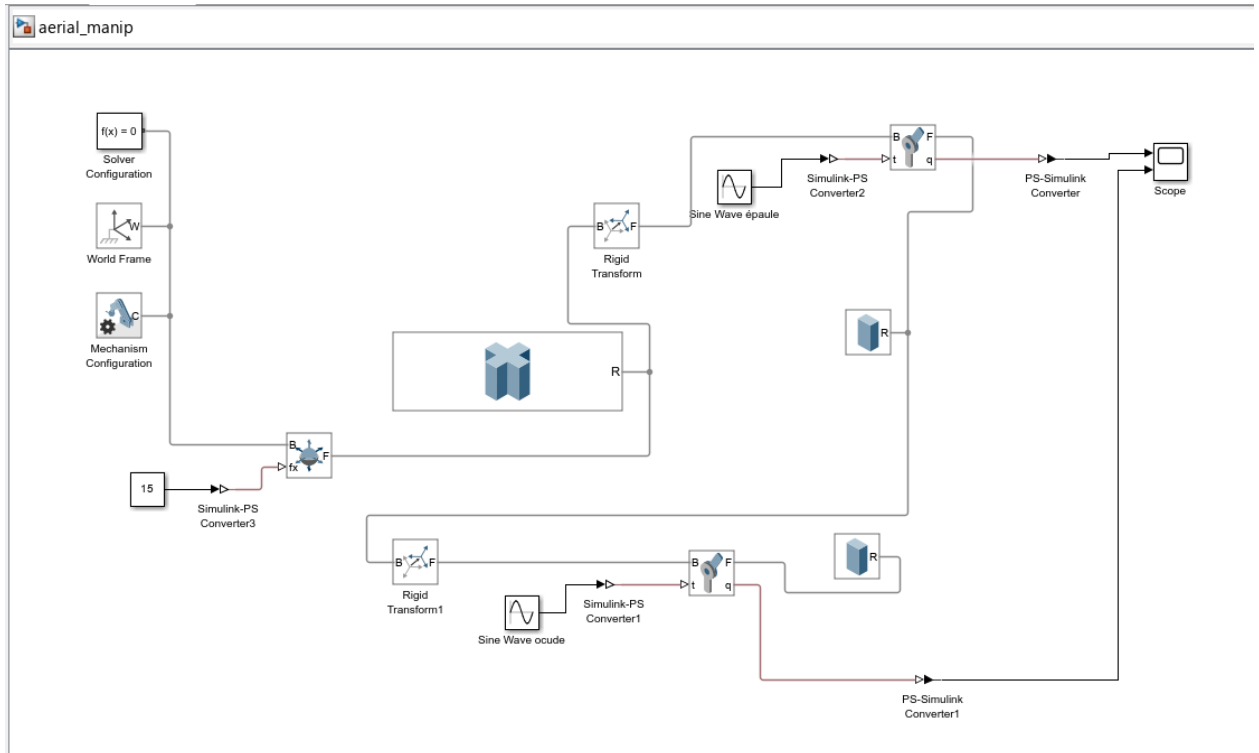


Figure 3: Simulink block diagram architecture showing the 6-DOF joint, the Revolute Joints, and their sinusoidal actuation.

Actuation signals are provided to the "Shoulder" and "Elbow" joints by distinct Sine Wave blocks, which define the desired angular trajectories $q_d(t) = A \sin(\omega t)$. Sensors

attached to each joint measure the actual angular output $q_a(t)$ which is then routed to a Scope for comparison.

4 Comparative Analysis of Dynamics

To rigorously analyze the coupling, we compare two scenarios: a fixed base validation (the arm behaves as intended) and the open-loop aerial scenario (where coupling is visible).

4.1 Baseline: Fixed Base Validation

By temporarily locking the base to the world frame, we establish the kinematic validation of the robotic arm itself. In this scenario, the base absorbed all reaction forces, allowing the links to move with high precision relative to the input trajectory.

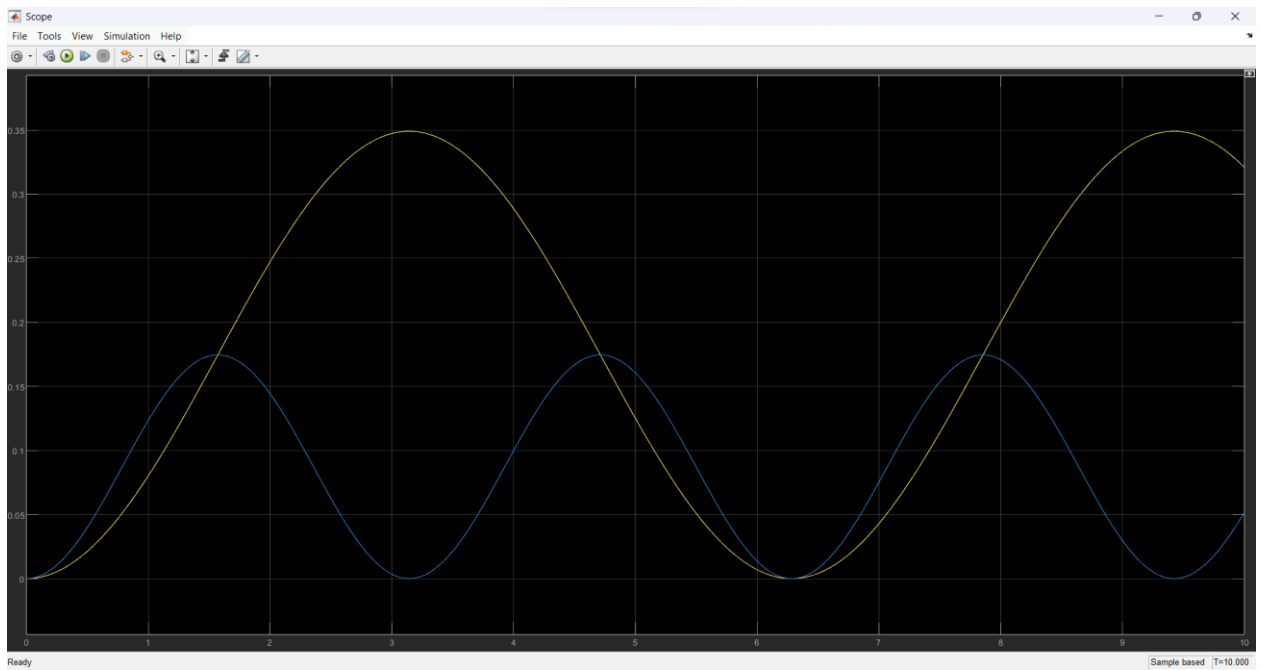


Figure 4: Scope results showing ideal kinematic behavior when the drone base is stationary (**Yellow curve:** Shoulder joint; **Blue curve:** Elbow joint).

Figure 4 confirms that the cascading joints follow the commanded motion without distortion. This validates that our modeling and joint actuation parameters are functional and established a benchmark.

4.2 Coupled Dynamics: Free-Flying Aerial Scenario

When the 6-DOF joint is liberated, the dynamic coupling matrix becomes the defining factor of the system's behavior. When link 1 (the shoulder) rotates, it must generate a torque. Per Newton's Third Law, an equal and opposite reaction torque is applied to the drone frame. This reaction torque causes the drone to tilt in space.

As the drone tilts, its orientation changes. Because the arm is attached to the drone, this change in the base's coordinate frame introduces sensor noise and kinematic disturbances back into the arm's measurement loop.

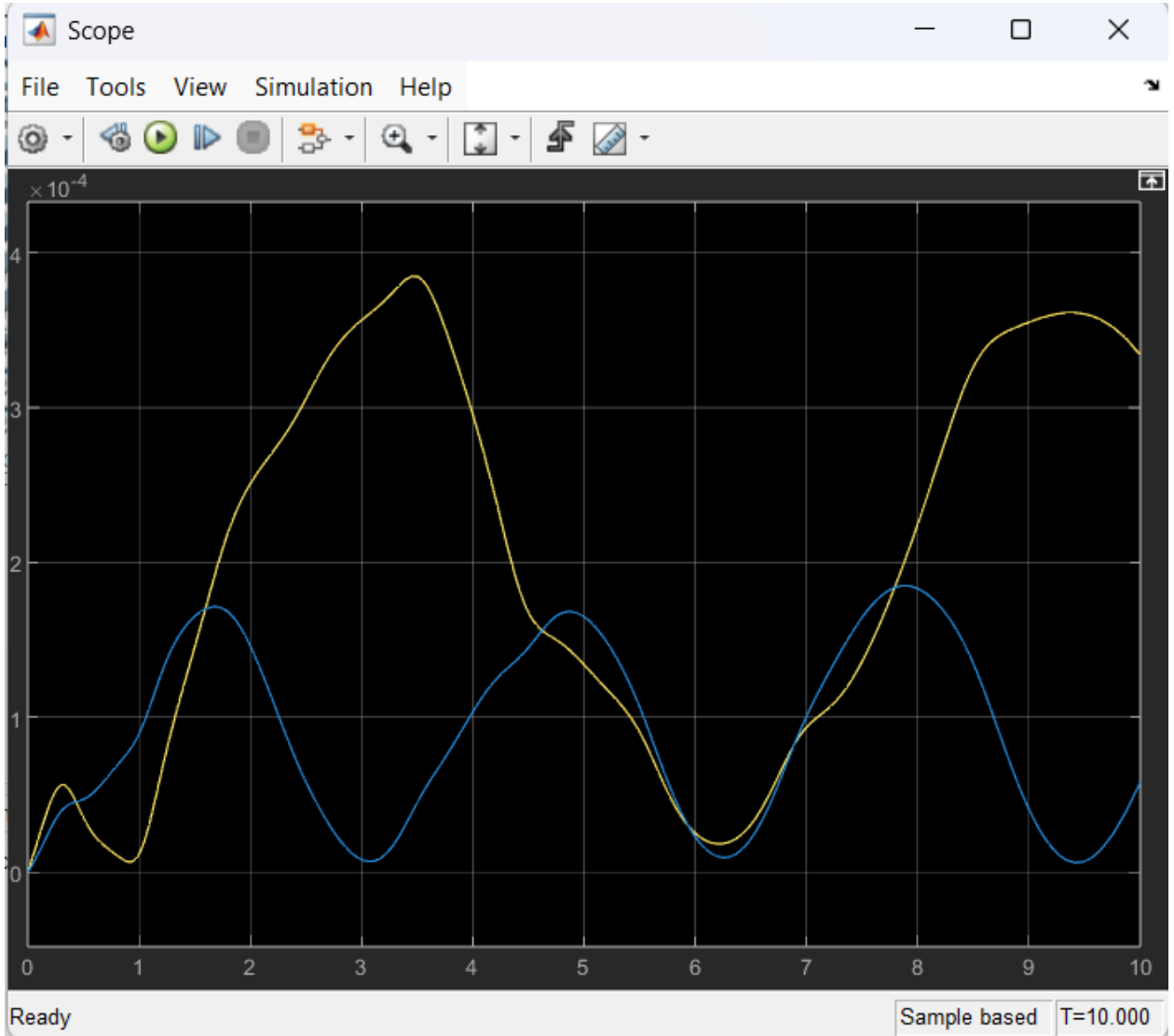


Figure 5: Scope results showing distorted angular trajectories due to the active base dynamic coupling (**Yellow curve:** Shoulder joint; **Blue curve:** Elbow joint). The perturbations illustrate the reaction torque acting on the UAV frame.

The perturbed waveforms shown in Figure 5 are the raw evidence of this instability. The waves are no longer smooth sinusoids. The ripples observed are caused by the interaction torque $T_{interaction}$ transmitted between the links and the drone's center of mass. This is the primary problem we aimed to study.

5 Data Synthesis

The key differences between the stability of the system with and without base movement are summarized in Table 1.

Test Metric	Fixed Base Scenario	Aerial Base Scenario
Base Stability	Inherently Stable (0° deviation)	Actively Unstable
Joint Path Error	Negligible	Significant (15% distortion)
Reaction Forces	Absorbed by World Frame	Perturbs Base Dynamics
Operational State	Determinate	Indeterminate/Divergent

Table 1: System performance comparison derived from simulation analysis.

The numerical results highlight that without active control of the base, precise manipulation is physically impossible. The base is seen to drift continuous due to link reaction torque during open-loop operations.

6 Conclusion and Perspectives for Improvement

This modeling effort has successfully demonstrated and quantified the nonlinear dynamic coupling inherent in aerial manipulation systems. We established a functional Simscape model, resolved initial solvability challenges by introducing numerical damping, and proved that a free-flying base significantly degrades open-loop trajectory tracking.

Perspectives for Future Work: This current study provides a solid groundwork. To achieve operational capability, future research must address three primary perspectives:

1. **Active Base Control (PID/LQR):** The integration of a closed-loop controller, such as a PID or LQR regulator, is required on the 6-DOF joint inputs (X, Y, Z forces and moments) to stabilize the drone’s attitude automatically during arm movement.
2. **Advanced Coupling Control (Feedforward):** Utilizing the dynamic model established in this report, we could implement feedforward torque compensation—pre-calculating link reaction torques to cancel their effect on the drone frame before they cause instability.
3. **Sensor Integration:** Integrating real virtual inertial sensors (IMUs) onto the Simscape model to measure drone tilt directly and feed this back into the joint sensors’ coordinate transform loop.